

DBMS Lab Practice Sheet

SQL: Joins, Aggregation & Subqueries

This sheet contains **questions derived from your faculty's materials** plus **additional challenging questions** commonly seen in DBMS lab finals. Each question shows only the expected output table — try writing the query yourself first!

■ Schema Reference

Employee (Ids, Name, Gender, Date_of_Birth, Date_of_Join, Permanent_Address, Current_Address, Phone, Type, Basic_Salary, Others, Dept_Id, Desig_Id)

Department (Dept_Id, Name) | **Designation** (Desig_Id, Name)

Manager (Ids, Name, Gender, Permanent_Address, Current_Address, Phone, Type, Managerial_Allowance, Dept_Id)

Teacher_Details (Ids, E_Mail, Papers) | **Officer_Details** (empty)

Section A — Joins (Faculty Material)

Q1. List all employee IDs from the Employee table.

Ids
1
2
3
4
5
6
7

Q2. Get the UNION of IDs from Employee and Manager tables (no duplicates).

Ids
1
2
3
4
5
6

Ids
7
8
9
10

Q3. Show a simple INNER JOIN (Cartesian product filtered) between Employee and Teacher_Details — display Name, Date_of_Birth, Date_of_Join, Phone, E_Mail, Papers.

Name	Date_of_Birth	Date_of_Join	Phone	E_Mail	Papers
Nasim	1979-12-17	2001-01-01	01730016854	nasim_1547@yahoo.co.uk	5
Arshad	1979-01-01	2005-01-01	01191540540	arshad_parvez@yahoo.com	4
Fahima	1975-01-01	1999-01-01	01819750750	fahima_noor@abc.ac.bd	2
Rafiq	1950-12-01	1991-01-01	01914684384	rafiq@abc.ac.bd	20
Salam	1970-04-15	1995-01-01	01715420420	salam@abc.ac.bd	9
Zia	1980-07-25	1999-01-01	01914789789	zia_arman@abc.ac.bd	2

Q4. Display Name, Department, Designation, E_Mail and Papers for all teachers (joining Employee, Department, Designation, Teacher_Details).

Name	Department	Designation	E_Mail	Papers
Nasim	CSE	Assistant Professor	nasim_1547@yahoo.co.uk	5
Arshad	BBA	Lecturer	arshad_parvez@yahoo.com	4
Fahima	ETE	Associate Professor	fahima_noor@abc.ac.bd	2
Rafiq	CSE	Professor	rafiq@abc.ac.bd	20
Salam	BBA	Associate Professor	salam@abc.ac.bd	9
Zia	ETE	Assistant Professor	zia_arman@abc.ac.bd	2

Q5. Show Name, Department, Designation, Total Salary (Basic+Others) for employees with Total Salary > 15000, ordered by Type descending.

Name	Department	Designation	Total_Salary	Type
Nasim	CSE	Assistant Professor	20000.00	T
Rafiq	CSE	Professor	60000.00	T
Salam	BBA	Associate Professor	25000.00	T
Zia	ETE	Assistant Professor	16700.00	T
Nowrin	CSE	Officer	12500.00	O

Q6. Show Name, Department, Designation, Basic_Salary for employees whose Basic Salary > 10000 AND name contains 'a', ordered by Basic_Salary ascending.

Name	Department	Designation	Basic_Salary
Nasim	CSE	Assistant Professor	15000.00

Name	Department	Designation	Basic_Salary
Fahima	ETE	Associate Professor	12000.00
Nowrin	CSE	Officer	10000.00
Salam	BBA	Associate Professor	20000.00

■ Note: Basic_Salary > 10000 AND name LIKE '%a%' — Arshad is excluded (Basic=12000 but 'Arshad' has 'a' — actually included; result ordered by Basic_Salary).

Q7. Perform a LEFT JOIN between Employee and Manager on Ids. Which employees appear even without a Manager match?

E.Ids	E.Name	E.Type	M.Ids	M.Name	M.Managerial_Allowance
1	Nasim	T	1	Nasim	20000.00
2	Arshad	T	2	Arshad	25000.00
3	Fahima	T	NULL	NULL	NULL
4	Nowrin	O	4	Nowrin	20000.00
5	Rafiq	T	NULL	NULL	NULL
6	Salam	T	NULL	NULL	NULL
7	Zia	T	NULL	NULL	NULL

→ Employees 3 (Fahima), 5 (Rafiq), 6 (Salam), 7 (Zia) have no Manager entry.

Q8. Perform a RIGHT JOIN between Employee and Manager on Ids. Which managers are NOT in the Employee table?

E.Ids	E.Name	M.Ids	M.Name	M.Managerial_Allowance
1	Nasim	1	Nasim	20000.00
2	Arshad	2	Arshad	25000.00
4	Nowrin	4	Nowrin	20000.00
NULL	NULL	8	Aslam	50000.00
NULL	NULL	9	Tasmia	70000.00
NULL	NULL	10	Samara	60000.00

→ Aslam (8), Tasmia (9), Samara (10) are managers not in the Employee table.

Section B — Aggregation (Faculty Material)

Q9. Count the total number of employees.

count(*)
7

Q10. Count the number of distinct departments in the Employee table.

count(distinct dept_id)
3

Q11. Find the maximum length of employee names.

max(length(name))
6

Q12. Find the minimum length of employee names.

min(length(name))
3

(Zia has 3 characters)

Q13. List distinct department IDs where Basic_Salary > 15000.

Dept_Id
1
2

(Rafiq:45000-CSE, Salam:20000-BBA)

Q14. Find the average Basic_Salary of all employees.

avg(Basic_salary)
18428.57

Q15. Find the average Basic_Salary for department 3 (ETE) only.

avg(Basic_salary)
13500.00

(Fahima:12000 + Zia:15000 = 27000 / 2 = 13500)

Q16. Show average Basic_Salary grouped by Dept_Id.

Dept_Id	avg(Basic_salary)
1	23333.33
2	16000.00
3	13500.00

(CSE: 15000+45000+10000=70000/3 | BBA: 12000+20000=32000/2 | ETE: 12000+15000=27000/2)

Q17. Show Department Name and average Basic_Salary per department (joining Employee and Department).

Name	avg(E.basic_salary)
CSE	23333.33
BBA	16000.00
ETE	13500.00

Q18. Show Department Name and average Total Income (Basic_Salary + Others) per department.

Name	avg(basic_salary+others)
CSE	27500.00
BBA	20000.00
ETE	16100.00

(CSE: (20000+13000+12500)/3 | BBA: (15000+25000)/2 | ETE: (15500+16700)/2)

Q19. Show Department Name and average Total Income for departments where avg(Basic+Others) > 15000. Use HAVING.

Name	avg(basic_salary+others)
CSE	27500.00
BBA	20000.00
ETE	16100.00

Q20. Show total Basic_Salary grouped by Gender and Type.

Gender	Type	sum(basic_salary)
F	O	10000.00
F	T	12000.00
M	T	107000.00

Q21. List all employees whose Basic_Salary is greater than the Basic_Salary of employee with Ids = 1 (Nasim, Basic = 15000). Use a subquery.

Ids	Name	Basic_Salary
5	Rafiq	45000.00
6	Salam	20000.00
7	Zia	15000.00

Note: Zia's salary equals 15000 — adjust to > to exclude or >= to include.

Q22. Find the highest average Basic_Salary among all departments. (Hint: use a nested subquery or derived table.)

MAX(avg_salary)
23333.33

Q23. Find the lowest average Basic_Salary among all departments.

MIN(avg_salary)
13500.00

Q24. Show Department Name and avg Total Income for departments where avg(Basic+Others) is greater than the overall average Basic_Salary.

Name	avg(basic_salary+others)
CSE	27500.00
BBA	20000.00
ETE	16100.00

Overall avg Basic_Salary $\approx$ 18428.57 — but all dept avg totals exceed it.

Section C — Extra Challenge Questions (Lab Final Style)

These questions go slightly beyond the notes but use only the concepts taught. They are typical of what appears in DBMS lab finals.

Q25. Find employees who are in the Employee table but NOT in the Manager table. (Use LEFT JOIN ... WHERE M.Ids IS NULL)

Ids	Name	Type
3	Fahima	T
5	Rafiq	T
6	Salam	T
7	Zia	T

Q26. Find managers who are NOT in the Employee table. (Use RIGHT JOIN ... WHERE E.Ids IS NULL)

M.Ids	M.Name	M.Managerial_Allowance
8	Aslam	50000.00
9	Tasmia	70000.00
10	Samara	60000.00

Q27. Show each department name along with the COUNT of employees in it.

Department	Employee_Count
CSE	3
BBA	2
ETE	2

Q28. List department names where the number of employees is more than 2.

Department	Employee_Count
CSE	3

Q29. Show Name and Total_Salary of the employee with the HIGHEST total salary (Basic_Salary + Others). Use a subquery.

Name	Total_Salary
Rafiq	60000.00

Q30. Find the name and department of employees who earn more than the average Basic_Salary of their own department. (Correlated subquery or self-join approach)

Name	Department	Basic_Salary	Dept_Avg
Nasim	CSE	15000.00	23333.33 — below avg
Rafiq	CSE	45000.00	23333.33 — ABOVE
Salam	BBA	20000.00	16000.00 — ABOVE

Expected result (only those ABOVE dept avg):

Name	Department	Basic_Salary
Rafiq	CSE	45000.00
Salam	BBA	20000.00
Zia	ETE	15000.00

Q31. Display Name, Designation, Department and Papers published, for teachers who have published MORE than 3 papers.

Name	Designation	Department	Papers
Nasim	Assistant Professor	CSE	5
Arshad	Lecturer	BBA	4
Rafiq	Professor	CSE	20
Salam	Associate Professor	BBA	9

Q32. Show the department that has the highest total (sum) of Basic_Salary. Use GROUP BY ... ORDER BY ... LIMIT 1.

Department	Total_Basic_Salary
CSE	70000.00

Q33. Using UNION, produce a combined list of all IDs from Employee and Manager (with duplicates removed), showing which IDs exist in BOTH tables.

(Hint: Use INTERSECT logic via INNER JOIN or subquery — MySQL doesn't have INTERSECT)

Common_Ids
1
2
4

Q34. Show the number of male and female employees separately.

Gender	Count
M	5
F	2

Q35. Find employees who joined BEFORE 2002, showing Name, Date_of_Join and Department Name.

Name	Date_of_Join	Department
Fahima	1999-01-01	ETE
Rafiq	1991-01-01	CSE
Salam	1995-01-01	BBA
Zia	1999-01-01	ETE

Name	Date_of_Join	Department
Nasim	2001-01-01	CSE

Q36. Display the department name(s) where the average total income (Basic+Others) is EQUAL TO the maximum average total income across all departments. (Nested subquery inside HAVING)

Department	Avg_Total_Income
CSE	27500.00

■ Tip: For questions involving INTERSECT or MINUS (not supported in MySQL), use INNER JOIN / LEFT JOIN ... IS NULL as equivalents.